Nile University

Nanoelectronics Integrated Systems Center (NISC)

Task 4

ECG Signal Preprocessing Pipeline for Time-Series Forecasting

Prepared by

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Introduction

This report outlines the robust preprocessing pipeline implemented for chaotic ECG time-series data, specifically targeting the MIT-BIH Arrhythmia database. The objective is to prepare the raw signals for Deep Learning forecasting models while strictly minimizing structural distortion to preserve non-linear dynamics.

Methodology and Execution

1. Baseline Wander Removal (Raw vs. Filtered)

Action: Extracted continuous data and applied non-linear median filtering.
Purpose: To acquire raw signals and immediately address low-frequency noise (e.g., patient movement). The median filter effectively centers the signal and removes baseline drift without distorting high-frequency R-peaks.

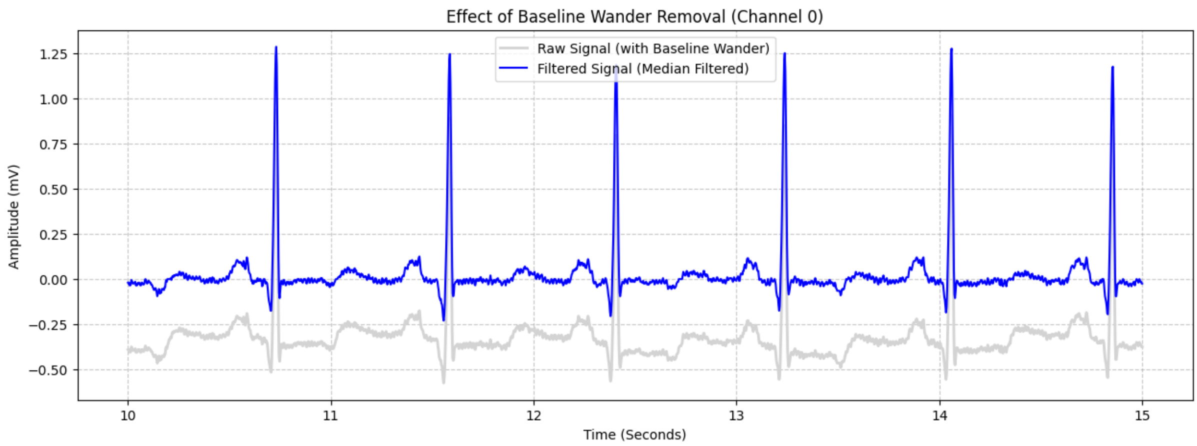

Figure 1: Effect of Baseline Wander Removal. The gray line represents the raw signal with baseline drift, while the blue line shows the centered, median-filtered signal.

2. Signal Denoising via Wavelet Transformation

Action: Applied Discrete Wavelet Transform (DWT) utilizing soft thresholding on detail coefficients.
Purpose: To eliminate high-frequency random noise (like muscle artifacts). DWT successfully thresholds this noise while preserving the sharp, vital features of the QRS complex, unlike traditional Fourier transforms which can compromise chaotic morphologies.

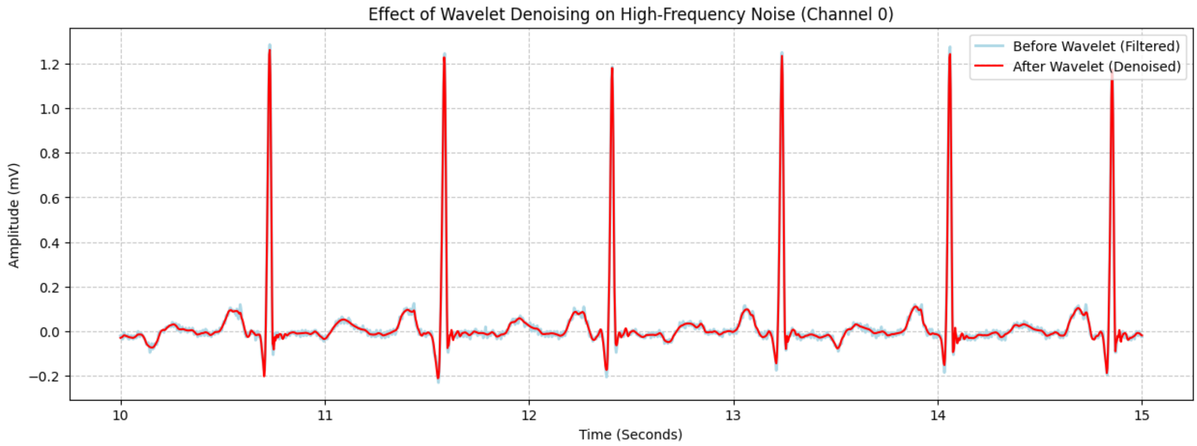

Figure 2: Effect of Wavelet Denoising. The light blue line is the signal before wavelet application, and the red line shows the smoothed, denoised signal with preserved QRS peaks.

3. Data Scaling and Forecasting Setup

Action: Implemented amplitude scaling and sliced the continuous data into sliding windows, mapping past sequences to future target sequences.
Purpose: Scaling ensures computational stability for Deep Learning models. Window-based slicing structures the dataset for causal forecasting, preserving the chronological integrity required to predict the future sequence (Target Y) based on the past sequence (Input X).

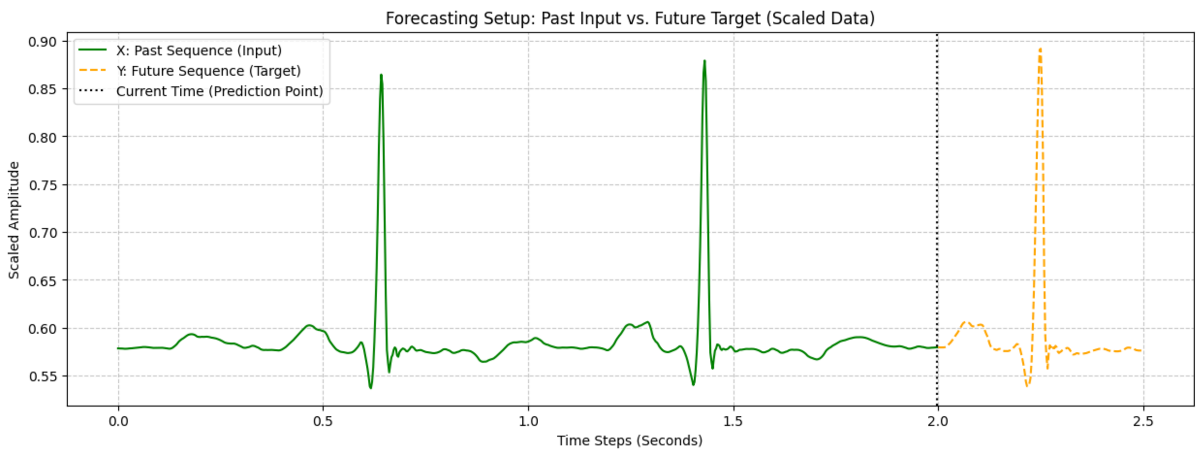

Figure 3: Forecasting Setup on Scaled Data. The green line represents the past input sequence (X), and the dashed orange line represents the future target sequence to be predicted (Y).

Conclusion

Visual inspection across all stages confirms the efficacy of the pipeline. Aggressive filtering was intentionally avoided, ensuring that the processed features fed into the deep learning model maintain their physiological and dynamic integrity, optimally formatted for time-series forecasting.